

Highlights

Modelling of Limit Order Book data with Ordered Fuzzy Numbers

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- The novel Limit Order Book data representation using the Ordered Fuzzy Numbers concept is presented.
- The proposed data representation stays mostly the same under data perturbation.
- The proposed and commonly used data representations are compared for predicting price movements under data perturbation.
- The proposed data representation maintains the performance of predicting models under data perturbation.

Modelling of Limit Order Book data with Ordered Fuzzy Numbers

Adam Marszałek^a, Tadeusz Burczyński^{a,b}

^a*Faculty of Computer Science and Telecommunications, Cracow University of Technology, 31-155, Cracow, Poland*

^b*Institute of Fundamental Technological Research, Polish Academy of Sciences, 02-106, Warsaw, Poland*

Abstract

This paper presents a novel approach to representing the Limit Order Book data at a given timestamp using the Ordered Fuzzy Numbers concept. The limit order book contains all buy and sell orders placed by investors, updated in real-time, for the most liquid securities, even several hundred times a minute. Due to its irregular nature (different and dynamic changes in the number of buy and sell orders), direct calculations on the order book data are not feasible without transforming it into feature vectors. Currently, most studies use a price level-based data representation scheme when applying deep learning models on limit order book data. However, this scheme has limitations, particularly its sensitivity to subtle perturbations that can negatively impact model performance. On the other hand, the ordered fuzzy number is a mathematical object (a pair of two functions) used to process imprecise and uncertain data. Ordered Fuzzy Numbers possess well-defined arithmetic properties. Converting the limit order book data to ordered fuzzy numbers allows the creation of a time series of ordered fuzzy numbers (order books) and use them for further calculations, e.g., to represent input data for deep learning models or employing the concept of fuzzy time series in various domains, such as defining liquidity measures based on limit order book data. In this paper, the proposed approach is tested using one-year market data from the Polish Stock Exchange for the five biggest companies. The DeepLOB model is employed to predict mid-price movement using different input data representations. The proposed representation of Limit Order Book data demonstrated remarkably stable out-of-sample prediction accuracy, even when subjected to data perturbation.

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1. Introduction

Financial market research constantly evolves by integrating innovative computational tools and novel mathematical concepts. The scope and form of data used in these studies are also evolving, from simple closing prices to candlestick chart data to complete order information from the limit order book. Today, more than half of the stock markets are order-driven markets that use electronic limit order books [1, 2]. Therefore, order limit book data is increasingly used as input to a wide range of computational models.

The Limit Order Book (LOB) is the central element of order-driven markets that records the intentions of buyers and sellers. It serves as a real-time representation of supply and demand for a particular financial instrument, displaying the changing price dynamics at a detailed, micro-structural level. LOBs, with their layers of price points and corresponding volumes, are inherently multi-dimensional, revealing complex spatial and time frames that explain price fluctuations. The structure of a LOB is complex, with multi-dimensional aspects resulting from multiple price points and order volumes spread over different levels for buy and sell orders. Figure 1 provides a snapshot of this dynamic system, illustrating the intricate interplay of buy and sell orders at different price levels.

To better understand the meaning of LOBs, it is necessary to highlight the wealth of data they generate. With the advent of algorithmic trading and the

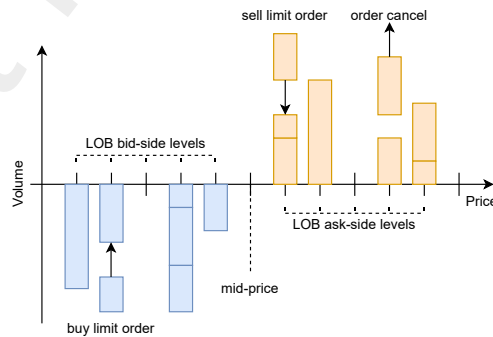


Figure 1: A snapshot of the limit order book.

digitisation of exchanges, a deluge of LOB data has spurred a renewed focus on data-driven approaches in financial markets. As a result, the intersection of machine learning, in particular deep learning and quantitative finance, has flourished [3, 4]. However, representing LOB data in machine learning models comes with its own set of challenges. While plenty of raw LOB data is available, translating this data into actionable insights requires sophisticated feature engineering or representation learning techniques. Interestingly, these advanced models are based on the initial representation of the LOB data. Currently, most studies use a price level-based data representation scheme. However, this representation has its limitations. Namely, it is susceptible to data perturbations. For example, traders are placing and canceling orders at deeper LOB levels, significantly amplifying the data's noise.

This paper addresses these difficulties by offering a fresh perspective on LOB data representation for machine learning models. Our goal is to challenge the dominant level-based representation of LOBs and propose a novel representation based on the concept of information modelling with ordered fuzzy numbers that capture the dynamic behaviour of LOBs over time while being robust to data perturbations due to the appearance and disappearance of small irrelevant orders.

The rest of the paper is organized as follows. In Section 2, the background and related research are provided. Section 3 describes limit order data and the standard representation scheme. A brief review of the foundational concept of ordered fuzzy numbers is presented in Section 4, followed by the introduction of a new method for representing Limit Order Book data. A range of experiments designed to test the robustness of the new LOB representation under different data perturbations is presented in Section 5. A benchmark dataset and model are also outlined in this section. A summary of the findings and considerations for future work and extensions are discussed in Section 6.

2. Literature Review and Related Work

The literature on machine learning models, in particular their application to financial market forecasting and high-frequency trading, is extensive and constantly evolving. Historically, the predictability of stock markets has been a significant area of interest in the economic literature. While there are differences of opinion on market efficiency, there is a consensus on market predictability to some extent [5, 6, 7, 8, 9, 10].

High-frequency trading (HFT) and stock market prediction have been recognized as complex areas, prompting interdisciplinary research approaches. For example, paper [11] used a logistic regression model to predict price jumps between trades, while paper [12] developed a multivariate linear model for short-term stock price movements. Also, neural networks have been highlighted for application, given the chaotic nature of the data [13]. Other studies have used various neural network architectures, such as RNNs and deep-belief networks, to predict currency rates, stock indexes, and design trading strategies [14, 15, 16, 17, 18, 19, 20, 21, 22].

Recently, there has been a marked interest in using machine learning algorithms to predict LOB data [23, 24, 25, 26, 27, 28, 29]. Given the stochastic nature of financial time series, these algorithms often involve preprocessing or feature extraction. Techniques such as principal component analysis (PCA) and linear discriminant analysis (LDA) have been implemented in this regard [25]. In particular, deep learning models have been widely applied to LOB data. Various model architectures, such as multilayer perceptrons [30], recursive neural networks [29], convolutional neural networks [31, 32], and self-attention transformer networks [33, 34] have been explored in research.

Machine learning models' effectiveness largely hinges on how data is represented. In neural networks, both the learning of representations and the forecasting steps are intertwined within the framework of the network and are collectively trained to optimize the same objective function. Here, the initial representation of LOB serves as the foundational input for the neural networks, shaping the entirety of the model. Current research primarily uses a price-level data representation scheme when applying deep learning models to LOB data [35, 26, 32, 30, 29, 28, 33, 34]. In the work [36], attention was drawn to the fact that there needs to be more literature regarding the compatibility of this representation scheme with deep learning models. This work showed that the price-level data representation scheme is sensitive to perturbations in the data, which leads to a decrease in the performance of advanced machine learning models.

3. Limit Order Book Data

The Limit Order Book is a dynamic record in the order-driven financial market that catalogues all current buy (bid) and sell (ask) orders that have been placed but not executed or cancelled. These orders are for specific assets and have a predetermined price and volume. LOB provides a comprehensive

overview of an asset's current market demand and supply. In electronic marketplaces, the LOB updates dynamically in real-time as new orders arrive, existing orders are cancelled or modified, or orders are executed [37].

The LOB is divided into two main sections. The bid side contains buy orders sorted in descending order based on their price, and the ask side contains sell orders sorted in ascending order based on their price. Each LOB side is divided into distinct levels based on submitted prices. Each level is characterised by price and volume of order.

Prices and volumes at different levels are often represented as vectors. The LOB vector representation helps capture the spatial structure and evolution of the market over time [23, 38, 36]. Spatially, price and volume at each level are interrelated. The spatial relationship at different levels is heterogeneous because there is no fixed gap between adjacent price levels.

For example, using a level-based representation, a LOB snapshot can be represented as a vector

$$s_t = \{p_a^i(t), v_a^i(t), p_b^i(t), v_b^i(t)\}_{i=1}^L, \quad (1)$$

where $p_a^i(t)$, $p_b^i(t)$ are the ask and bid prices for price level i at time t and $v_a^i(t)$, $v_b^i(t)$ are the ask and bid volumes, respectively. The time sequences of these snapshots represent the evolution of the market, and this data structure can be represented as $S \in \mathbb{R}^{T \times 4L}$, where T is the history length and L is the number of (nonzero) price levels considered for each side.

This vector structure, while effective and intuitive from the point of view of human understanding and compatible with the matching engines in exchanges, has several disadvantages from the point of view of input data for machine learning models [36]. The spatial structure at different levels is not homogeneous because it is not assumed that adjacent price levels have constant spacing, which is the basic assumption for convolutional neural networks (CNNs) due to the parameter-sharing mechanism. In addition, this representation can sometimes be unstable due to dynamic changes in price levels, e.g., the previous best bid/ask data may suddenly move to second place when a new order with a better price is placed. Similarly, the appearance of a relatively small order can dramatically change the norm of this vector. Moreover, only the first L levels (usually 10) are considered in this representation, so these levels may take orders with a small volume while kicking out levels with a much larger volume.

4. Proposed representation of LOB

4.1. Why Ordered Fuzzy Numbers?

Ordered fuzzy numbers (OFNs) are a specific type of fuzzy numbers introduced by Kosinski et al. (also called Kosin'ski's fuzzy numbers) in the series of papers [39, 40, 41, 42, 43]. OFNs, in contrast to classical fuzzy numbers [44, 45], are defined by ordered pairs of continuous real functions defined on the interval $[0, 1]$, i.e.,

$$A = (f, g) \text{ with } f, g: [0, 1] \rightarrow \mathbb{R} \text{ as continuous functions,} \quad (2)$$

instead of the classical membership function $\mu_A: \mathbb{R} \rightarrow [0, 1]$. The ordered fuzzy number and the ordered fuzzy number as a fuzzy number in classical meaning are presented in Fig. 2

The main advantage of OFNs is that the basic arithmetic operations $\{+, -, \cdot, \div\}$ are defined as the pairwise operations of their elements, which leads to the existence of the neutral elements of addition and multiplication. This fact means that the result of an arithmetic operation is not always a fuzzy number with larger support. Thanks to this, we can build fuzzy models based on ordered fuzzy numbers in the form of classical equations without losing accuracy. Essential mathematical functions such as *log*, *exp*, *sqrt*, and more can be defined in a similar way (see [46]). Moreover, an OFN is a generalization of the real number in the sense that each real number a can be represented as the OFN with constant functions f and g equal to a .

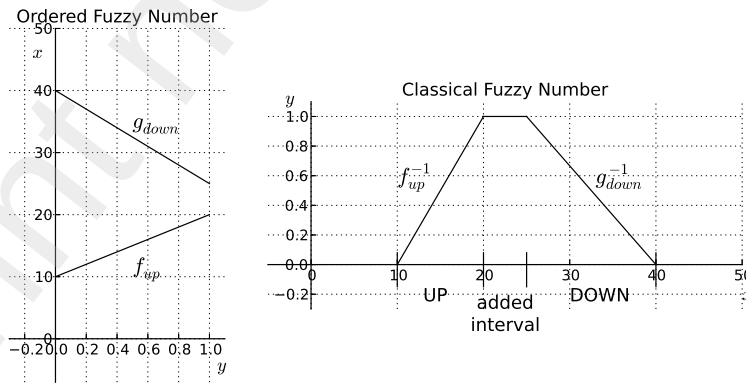


Figure 2: Graphical interpretation of OFN and a OFN presented as fuzzy number in classical meaning [47].

Ordered fuzzy numbers have found many applications in various areas, such as modelling dynamic changes, decision-making, DDoS attack analysis, and many others [48, 49, 50, 51, 52]. In particular, in our previous works [53, 54, 47, 55], they have been used to model/represent high-frequency stock market data in the form of a well-defined mathematical object, based on which it becomes possible to build more robust and flexible models that better reflect the complexity and uncertainty characteristic of financial markets.

Figure 3 shows a graphical representation of the LOB snapshot (real data, KGHM on 2017-01-03 09:22:52.299826, only price levels in range $\pm 100\%$ of mid-price) and an illustration of the corresponding demand and supply (the number of shares we can sell or buy at a fixed price limit) for the same limit order book data, respectively. We note that the demand and supply lines can be identified with functions f and g . However, to be an OFN, they must be functions defined on the interval $[0, 1]$. For this purpose, we developed a transformation method, which is presented in the section below.

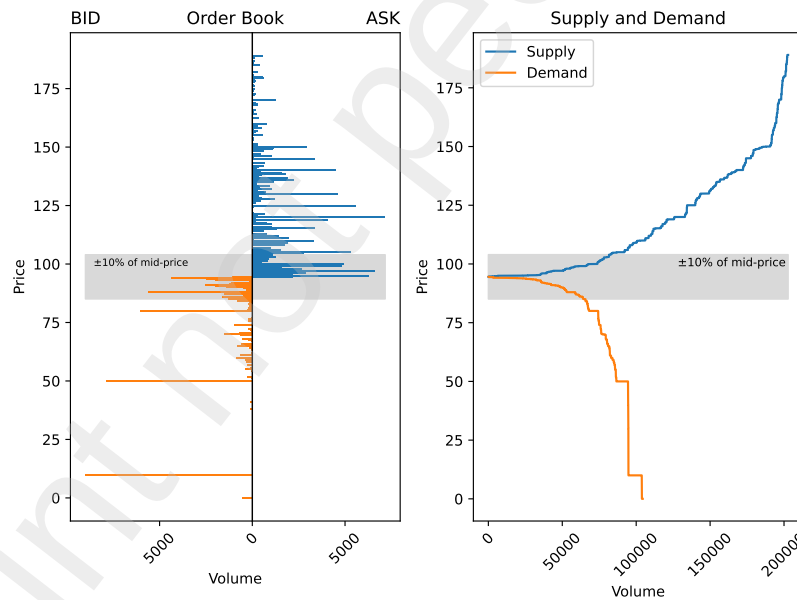


Figure 3: Limit order book snapshot for KGHM on 2017-01-03 09:22:52.299826 and the corresponding demand and supply chart.

4.2. Limit Order Book as Ordered Fuzzy Number

One of the interesting areas of the limit order book analysis is modelling and measuring liquidity. The ability to find and estimate intraday liquidity quickly and accurately is extremely valuable but also very challenging. A perfectly liquid market is one in which any amount of a given security can be instantaneously converted into cash and back to securities at no cost. In the real world, a liquid market is one where the transaction costs associated with this conversion are minimised [56]. Market liquidity is not unidimensional and can be understood in the following aspects [57]:

- the quantity of securities that are traded (depth)
- the ability of the security prices to quickly recover after a liquidity shock (resiliency)
- the costs incurred in trading security (tightness)
- the time taken to execute a trade (immediacy)
- the intensity of trading volume impact on security prices (breadth)

In this paper, we propose conversion of LOB to OFN in such a way that OFN, in a simple way, can illustrate the depth of a given instrument, i.e., it shows how many shares we are able to sell or buy at a given moment and how it will affect the share price.

Let $\{p_a^i(t), v_a^i(t)\}_{i=1}^{L_a}$ and $\{p_b^i(t), v_b^i(t)\}_{i=1}^{L_b}$ be a complete snapshot of LOB at time t (all price levels), where $p_a^i(t)$, $p_b^i(t)$ are the ask and bid prices for price level i at time t and $v_a^i(t)$, $v_b^i(t)$ are the ask and bid volumes, respectively, and L_a , L_b are the number of (nonzero) price levels considered for ask and bid side, respectively. Moreover, let $p_r(t)$ be a reference price at time t (e.g. mid-price or open price). Then the function f and g of OFN $A_t = (f_t, g_t)$ at time t are defined as follows:

$$f_t(x) = - \begin{cases} 0 & \text{if } (1-x)p_r(t) > \mu_a^1(t), \\ \sum_{i=1}^{L_a} v_a^i(t) & \text{if } (1-x)p_r(t) \leq \mu_a^{L_a}(t), \\ \left(\sum_{i=1}^{l_a} v_a^i(t) \right) \left(1 + \frac{\mu_a^{l_a}(t) - (1-x)p_r(t)}{(1-x)p_r(t) - p_a^{l_a+1}(t)} \right) & \text{otherwise} \end{cases} \quad (3)$$

$$g_t(x) = \begin{cases} 0 & \text{if } (1+x)p_r(t) < \mu_b^1(t), \\ \sum_{i=1}^{L_b} v_b^i(t) & \text{if } (1+x)p_r(t) \geq \mu_b^{L_b}(t), \\ \left(\sum_{i=1}^{l_b} v_b^i(t) \right) \left(1 + \frac{\mu_b^{l_b}(t) - (1+x)p_r(t)}{(1+x)p_r(t) - p_b^{l_b+1}(t)} \right) & \text{otherwise} \end{cases} \quad (4)$$

where $\mu_a^i(t)$ and $\mu_b^i(t)$ are volume-weighted average prices at time t from level 1 to i for ask and bid prices, respectively, l_a is the lowest i that satisfies the relation $\mu_a^i < (1-x)p_r(t)$ and l_b is the lowest i that satisfies the relation $\mu_b^i > (1+x)p_r(t)$.

Interpretation

Figure 4 shows an example of an ordered fuzzy number generated by the limit order book of KGHM on 2017-01-03 09:22:52.299826. The values of $|f(x)|$ and $g(x)$ show the size of trades that can be made for a given level of potential cost $(1-x)$, i.e., the percentage difference between the real average transaction price and the reference price for the ask and bid side, respectively. For example, if an investor wishes to purchase 30,000 shares of KGHM all at once, the average cost per share would be 94.97. Compared to the current mid-price of 94.52, this would result in an approximate loss of 0.4% if the investor could have bought the shares at the mid-price. Likewise, if the investor wants to sell 10,000 shares right now, the average selling price per share would be 94.22. This would lead to an approximate loss of 0.3% compared to the current mid-price.

Additional parameters and modifications for practical use cases

The general definition of an ordered fuzzy number generated by limit order book data has been defined above. In order to better adapt it to specific practical applications, we propose considering a few additional parameters and modifications described below.

Numbers of price levels. Considering all orders in the limit order book can be problematic and often inappropriate, there may be orders that cannot be executed in any given session due to price fluctuation constraints. Therefore, we propose to take into account only price levels within specific reference price bounds ($\pm$ daily price fluctuation limit, e.g., $\pm 10\%$). The result of applying this restriction is shown in Figure 5.

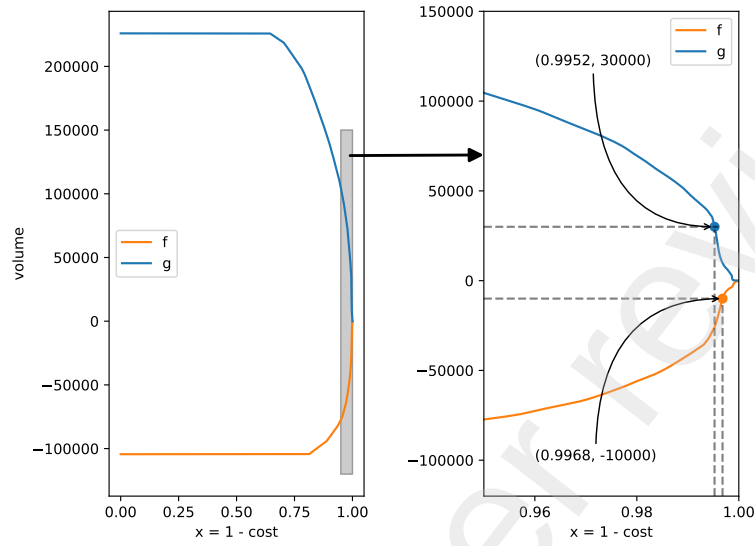


Figure 4: The ordered fuzzy number generated by limit order book of KGHM on 2017-01-03 09:22:52.299826.

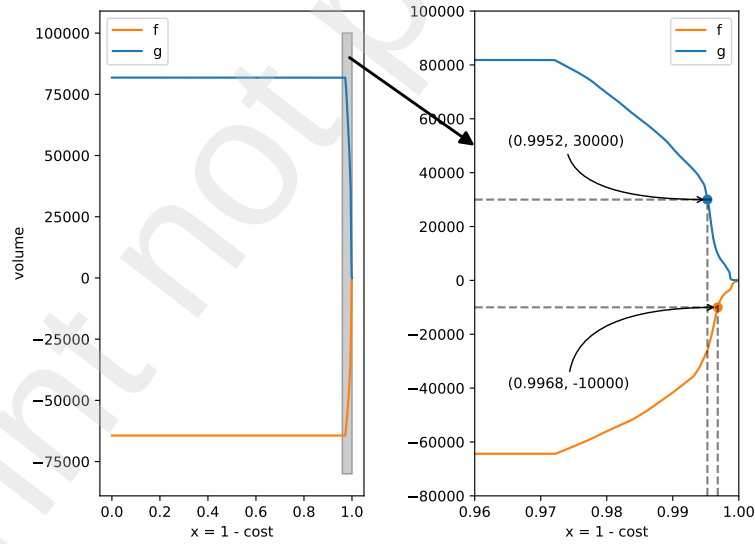


Figure 5: The ordered fuzzy number generated by limit order book of KGHM on 2017-01-03 09:22:52.299826 under restriction to price levels in range of $\pm 10\%$ of mid-price.

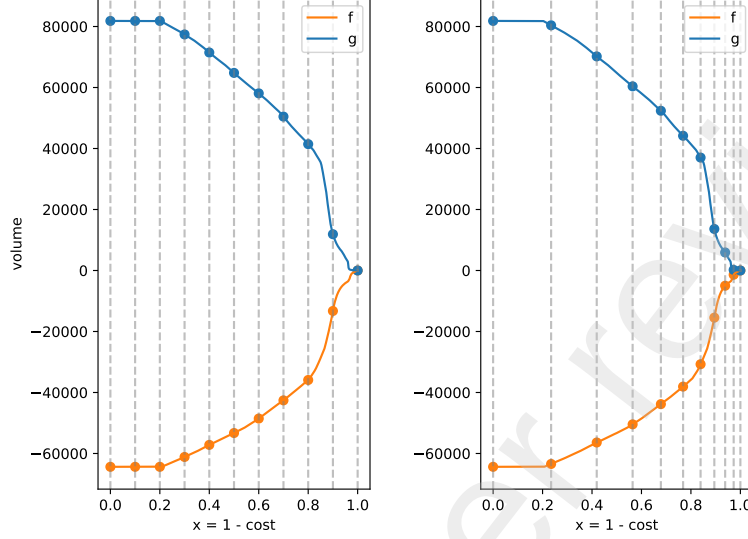


Figure 6: The ordered fuzzy number generated by the limit order book of KGHM on 2017-01-03 09:22:52.299826 under restriction to price levels in the range of $\pm 10\%$ of mid-price and maximum cost of 3.5%. The labelled markers indicate the distribution of points when using linear and geometric progressions for discretisation, respectively.

Maximum cost. Figure 5 also illustrates that the bulk of the pertinent data is concentrated close to $x=1$. This concentration stems from the limited scope of examined price levels and the method of computing cost as a percentage difference between the average transaction price and the reference price. To address this, we recommend implementing a parameter that sets the upper limit for the considered cost (e.g., single transaction price fluctuation limit, e.g., 3.5%). Subsequently, adjust the x -axis linearly so that $x=0$ aligns with the maximum predetermined cost and $x=1$ aligns with the zero cost. The result of this transformation for the assumed maximum cost of 3.5% is shown in Figure 6.

Discretisation. In prior research, for numerical calculations, we have represented an ordered fuzzy number $A = (f, g)$ as an ordered pair of vectors $(\mathbf{f}, \mathbf{g})$. These vectors $\mathbf{f}$ and $\mathbf{g}$ are constructed from the function values of f and g , respectively. These values are calculated at $(M + 1)$ points, which are obtained by uniformly discretising the interval $[0, 1]$, i.e.

$$\begin{aligned} \mathbf{f} &= [f(0), f(dx), f(2 \cdot dx), \dots, f((M - 1) \cdot dx), f(1)], \\ \mathbf{g} &= [g(0), g(dx), g(2 \cdot dx), \dots, g((M - 1) \cdot dx), g(1)], \end{aligned} \quad dx = \frac{1}{M}. \quad (5)$$

This numerical representation of OFN allows for convenient implementation of arithmetic operations, resulting in ordered fuzzy numbers with functions f and g of any shape. In the case of LOB data representation, we propose replacing the linear scale with a logarithmic scale (a geometric progression) to increase the density of points close to $x=1$ (see Figure 6).

5. Experiments and Results

5.1. Benchmark Dataset and Model

A commonly used LOB dataset for model testing is the FI-2010 dataset [23], which is the first publicly available benchmark dataset of high-frequency limit order book data. This dataset contains time series data for five Nasdaq Nordic stock market stocks for ten consecutive days. However, this publicly available data is based on only ten price levels and has already been pre-processed (normalised). Therefore, it is impossible to prepare input data for deep learning models based on them using the LOB data representation method proposed in this paper. Moreover, as mentioned in [32], a 10-day dataset is inadequate for thoroughly evaluating an algorithm's robustness and capacity for generalisation, given that the issue of overfitting to historical data is significant and a signal is typically expected to remain stable over several months. That is why they also trained and tested their model on their dataset of limit order book data of one-year length for instruments that are among the most liquid stocks listed on the London Stock Exchange. Unfortunately, this dataset is not publicly available.

We are deeply grateful for the financial support from The National Science Centre, which enabled us to acquire invaluable historical data from the Warsaw Stock Exchange (WSE), encompassing details on all orders and transactions. Additionally, with the consultation and approval of the WSE, we have prepared an open-access version of our data to share with the broader research community. A brief description of the data is provided below, while more details were released in the related article in *Data in Brief* journal.

WSE LOB dataset. We have compiled a year-long dataset for five key stocks: KGHM, PKNORLEN, PKOBP, PZU, and PEKAO. These are among the most liquid stocks on the Warsaw Stock Exchange in 2017. Our dataset covers all LOB updates (messages) and spans every trading day from January 1, 2017, to December 24, 2017. From the raw message books, one can reconstruct complete limit order books. For the purposes of our research, we

generated time series data of the LOB, which, for each state, includes only the price levels that fall within the daily price fluctuation limit of $\pm 10\%$. This limit is calculated based on the actual mid-price rather than the reference price. Each level in the series includes both price and volume information. We have limited the timeframe to between 9:00 AM and 4:50 PM (CET or CEST, as applicable), ensuring that only standard trading activities are included and auctions are excluded. Also, we omit timestamps where all selected levels remain unchanged. In summary, our dataset from the Warsaw Stock Exchange spans a year and includes almost 79 million data points. Each stock experiences, on average, about 63,000 events per day, and these events occur at inconsistent time intervals. The time gap between any two successive events can range from a fraction of a second to several minutes. The average time gap in the dataset is 0.481713 seconds. Statistics descriptions for individual stocks are presented in Table 1. For analysis, we use the initial nine months of data for training (with the separation of the last 10% of the records as validation data) and the final three months for testing. Given the high-frequency nature of the data, the 3-month test period includes millions of data points, making it adequate for evaluating the model's effectiveness and accuracy.

Table 1: Statistical description of the dataset segmented by individual stocks.

Stock	LOB snapshots	LOB snapshots per day	Mean Gap
KGHM	19,500,582	78,002.328	0.361515s
PKNORLEN	18,261,547	73,046.188	0.386032s
PKOBP	16,322,175	65,288.7	0.431902s
PZU	15,743,871	62,975.484	0.447772s
PEKAO	9,022,376	36,089.504	0.781342s

DeepLOB model. For the benchmark model, we selected the DeepLOB model [32]. This model is notable for being the first hybrid deep neural network designed to predict stock price movements using high-frequency limit order book data. It combines convolutional layers with Long Short-Term Memory (LSTM) units to achieve superior predictive performance compared to other existing algorithms that also use LOB data for feature extraction as of the time it was published. The authors of the DeepLOB model validated its

performance using two datasets. The first was the FI-2010 benchmark LOB dataset, and the second was a comprehensive, year-long dataset featuring 134 million data points from the London Stock Exchange. We chose to use only the DeepLOB model for our experiments due to computational constraints, which include the need to train the model multiple times for various assets and input data types. Another reason for our choice was the similarity between the London Stock Exchange dataset used in the original DeepLOB model and our own dataset.

5.2. Inputs, Normalisation and Labelling

To evaluate the effectiveness of the new LOB data representation method, we assembled two distinct sets of input data for each asset.

The first set employs the traditional level-based LOB representation, as outlined in Section 3. In this method, each LOB state includes ten levels on both the bid and ask sides, featuring both price and volume data. This results in 40 features per timestamp. We use the 50 most recent LOB states for modelling purposes as input. In particular, a single input is defined as

$$X^{(1)} = [x_1^{(1)}, x_2^{(1)}, \dots, x_t^{(1)}, \dots, x_{50}^{(1)}]^T \in \mathbb{R}^{50 \times 40}, \quad (6)$$

where

$$x_t^{(1)} = [p_a^i(t), v_a^i(t), p_b^i(t), v_b^i(t)]_{i=1}^{10}, \quad (7)$$

and $p_a^i(t)$, $p_b^i(t)$ are the ask and bid prices for price level i at time t and $v_a^i(t)$, $v_b^i(t)$ are the ask and bid volumes, respectively.

The second set comprises LOB snapshots formulated using the OFN approach, detailed in Section 4. In this case, a single input is designated as

$$X^{(2)} = [x_1^{(2)}, x_2^{(2)}, \dots, x_t^{(2)}, \dots, x_{50}^{(2)}]^T \in \mathbb{R}^{50 \times 40}, \quad (8)$$

where

$$x_t^{(2)} = [(1 - r_i r_{max})p_r(t), f_t(r_i)/10^6, (1 + r_i r_{max})p_r(t), g_t(r_i)/10^6]_{i=1}^{10}, \quad (9)$$

where $A_t = (f_t, g_t)$ is the ordered fuzzy numbers generated by LOB data at time t , p_r is the reference price calculated as mid-price at time t , $r_{max} = 0.035$ is the maximum considered cost, and r_i are 10 points from the interval $(0, 1]$ spaced evenly on a log scale (a geometric progression, i.e. 0.02709816, 0.06153943, 0.10531364, 0.16094986, 0.23166248, 0.32153691, 0.43576567, 0.58094831, 0.76547279, 1).

In the same manner as previous studies [23, 32, 36], we employed z-score standardisation to normalise our data for both types of inputs. We used the mean and standard deviation from the preceding five days to normalise each day's data, applying this separately for each financial instrument. A static normalisation approach is unsuitable for a one-year dataset because financial time series often undergo changing trends. This dynamic method ensures that the normalised data generally falls within a reasonable range.

Our goal is to predict the micro-movements of the mid-price, which is defined as

$$p_t = \frac{p_a^{(1)}(t) + p_b^{(1)}(t)}{2}. \quad (10)$$

Directly comparing the mid-prices p_t and p_{t+k} can result in a noisy label set. To address this, we employ a smoothing labelling method similar to one used in previous research [32, 23, 27]. Let m_- denote the mean of the previous k mid-prices and m_+ denote the mean of the next k mid-prices

$$m_-(t) = \frac{1}{k} \sum_{i=0}^k p_{t-i}, \quad (11)$$

$$m_+(t) = \frac{1}{k} \sum_{i=0}^k p_{t+i}, \quad (12)$$

where p_t is the mid-price defined in Equation (10) and k is the prediction horizon. We calculate the percentage change l_t in the mid-price to determine its direction, using a following equation

$$l_t = \frac{m_+(t) - m_-(t)}{m_-(t)}. \quad (13)$$

We then categorize the movement into three classes: -1 for downward movement, 0 for stationary, and $+1$ for upward movement. The final labels are determined based on a threshold α_k for the percentage change l_t . If $l_t > \alpha_k$ or $l_t < -\alpha_k$, the movement is classified as upward ($+1$) or downward (-1), respectively. For all other cases, it is considered stationary (0). Table 2 lists the threshold values α_k for specific prediction horizons k . These thresholds were chosen to ensure a balanced distribution of classes in the training dataset.

Table 2: The values of threshold α_k for specific prediction horizons k for individual stocks.

Stock \ k	20	50	100
KGHM	0.0000263	0.000102	0.000155
PKNORLEN	0.000022	0.000085	0.000145
PKOBP	0.0000207	0.000089	0.000147
PZU	0.000029	0.0000845	0.000136
PEKAO	0.00005	0.00012	0.000185

5.3. Data perturbation

We employed a straightforward data perturbation technique inspired by the method introduced in [36] to assess the stability of our proposed method of LOB data representation. In specific LOB data related to stocks, the price gap between adjacent levels sometimes exceeds the tick size (the smallest allowable price change). This phenomenon is notably common in small-tick stocks and can cause a complete shift in the LOB even with placing a minimum-size order at a price between existing levels. This perturbation method assumes that small orders are placed at vacant price levels beyond the best ask/bid prices, ensuring that the mid-price remains unchanged, thereby keeping prediction labels intact. We demonstrate this technique using a real LOB example. The first part (A) of Figure 7 shows a LOB snapshot with ten price levels on each of the ask and bid sides (labeled L1-L10) before any perturbation. The mid-price in this snapshot is 93.93, with a bid-ask spread of 0.04. Assuming a tick size of 0.01 and a minimum order size of 1, some price levels, like 93.96 and 93.97 on the ask side and 93.87 and 93.86 on the bid side, are empty. To perturb this data, one can place minimum-size orders at these empty levels, which appears inconsequential as they neither affect the mid-price nor add significant volume.

Upon perturbation, nearly half of the original price level details become obscured (see Fig. 7 (B)). For instance, ask-side levels L6 to L10 vanish post-perturbation, while the remaining levels shift in their LOB positioning. This perturbation has two critical effects from a machine-learning standpoint. First, it dramatically alters the 40-dimensional input space; for example, the Euclidean distance between the 40-dimensional vectors pre and post-perturbation is 6129.09 (0.22 when the volume is normalised by a factor of 10^6), even though the total order volume applied is merely 8. Second, it limits the model's ability to "see" the market, as represented by the grey

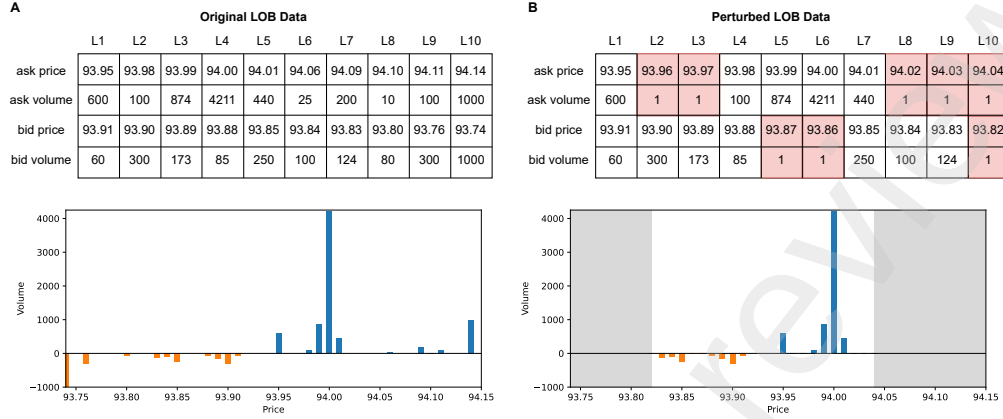


Figure 7: (A) Original LOB data with 10 levels on ask and bid side without perturbation. (B) LOB data with 10 levels after data perturbation.

areas masked out in the LOB visualization after perturbation.

It is worth noting that in the proposed representation method, the number of price levels considered is variable, depending on the price range. Even though we consider more price levels, we perturb only between L1 and L10 to maintain the same level of disruption. As a result, this method is highly resilient to such perturbations because new orders will remove none of the existing levels. Moreover, the impact of small orders is minimal. For instance, the Euclidean distance between the 40-dimensional vectors generated by our method before and after perturbation is 0.02 (1464.44 when the cumulative volume, i.e., values of functions f and g are not normalise by a factor of 10^6).

Note that in the data perturbation method outlined above, all available gaps are filled with new small orders. To add flexibility to this approach, we introduce an extra parameter, P , which indicates the percentage of empty price levels that will be filled with these orders. Our empirical studies will explore various perturbation levels, specifically 0% (no perturbation), 1%, 5%, 10%, 25%, and 50%.

5.4. Experiments Settings

We apply the publicly accessible version of the DeepLOB model¹ for all our experiments. We adopt the same settings for training of weights as outlined in the original study, i.e., using categorical cross-entropy loss and the ADAM optimisation algorithm. The learning is stopped if there is no improvement in validation accuracy over 20 consecutive epochs, with a maximum of 50 epochs. Our training employs mini-batches of 64 samples, as suggested in [32, 58]. All models are built using Keras [59] based on the TensorFlow backend [60], and we train them using a mirrored strategy of distributed learning on computing machines equipped with two NVIDIA Tesla V100 GPUs.

We trained models separately for each of the five assets (KGHM, PKNORLEN, PKOBP, PZU, and PEKAO) in two variants of input data (standard and proposed) and for three forecast horizons ($k \in \{20, 50, 100\}$), which gave us $5 \times 2 \times 3 = 30$ different configurations for training. Each model is trained with a non-perturbed training set, and when testing the model, we apply data perturbation with various levels of perturbation $P \in \{0\%, 1\%, 5\%, 10\%, 25\%, 50\%\}$. In addition, all models were re-trained from scratch five times each for added statistical robustness of results.

5.5. Model Performance

Table 3 showcases the testing efficacy of the DeepLOB model in predicting price movements within the WSE dataset. The results are averaged across five stocks, with individual results for each stock detailed in Appendix A. The table explores variations in perturbation levels (P), prediction horizons (k), and input types (level-based $X^{(1)}$ and the newly proposed $X^{(2)}$). We evaluate the model's performance using four distinct metrics: Accuracy (%), Precision (%), Recall (%), and F-score (%). Accuracy (%) is calculated as the percentage of test sample predictions that precisely align with the actual outcomes. This is considered an unbalanced accuracy score, while the remaining metrics are weighted averages across different classes. We display the confusion matrices in Figure 8 to better investigate the results.

First, in a disturbance-free scenario ($P = 0\%$), the DeepLOB model shows performance comparable to that outlined in article [32] for the LSE

¹<https://github.com/zcakhaa/DeepLOB-Deep-Convolutional-Neural-Networks-for-Limit-Order-Books>

Table 3: Experiment results on WSE dataset for various of perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	64.53	65.45	67.1	67.5	64.53	65.45	64.9	65.76
	50	62.11	62.56	61.57	62.31	62.11	62.56	61.77	62.42
	100	61.57	62.06	61.04	61.49	61.57	62.06	61.24	61.69
1%	20	55.05	65.22	54.79	67.16	55.05	65.22	54.57	65.53
	50	55.27	62.41	54.27	62.21	55.27	62.41	53.34	62.3
	100	54.81	62.01	53.64	61.45	54.81	62.01	53.38	61.64
5%	20	47.84	64.36	48.78	65.9	47.84	64.36	46.2	64.65
	50	47.85	61.78	48.59	61.82	47.85	61.78	44.44	61.8
	100	48.07	61.74	47.99	61.23	48.07	61.74	45.23	61.42
10%	20	45.21	63.5	46.82	64.72	45.21	63.5	42.65	63.77
	50	45.4	61.1	46.7	61.42	45.4	61.1	41.26	61.24
	100	45.74	61.42	46.03	60.97	45.74	61.42	42.4	61.15
25%	20	42.49	61.91	44.77	62.69	42.49	61.91	38.82	62.12
	50	42.71	59.44	44.43	60.54	42.71	59.44	37.9	59.84
	100	42.83	60.61	43.48	60.4	42.83	60.61	38.87	60.49
50%	20	41.7	61.14	43.99	61.78	41.7	61.14	37.99	61.33
	50	41.58	58.42	42.97	60.03	41.58	58.42	36.93	58.94
	100	41.39	60.02	42.02	60.05	41.39	60.02	37.49	60.03

dataset. Like the findings there, the model’s effectiveness declines as the prediction horizon expands. While the impact of changing input types is minor, the proposed input method slightly outperforms all metrics across various forecasting horizons. This suggests that our proposed input representation method captures critical information about the limit order book as effectively as traditional level-based representation schemes.

Second, in line with the observations made in the work [36], the model’s performance deteriorates when unexpected data distortions occur, particularly with the traditional input representation. Compared to a perturbation-free environment, we notice an accuracy drop ranging from approximately 7% ($P = 1\%$) to even 23% ($P = 50\%$). The confusion matrix data further indicates that, under the traditional input scheme, the model increasingly

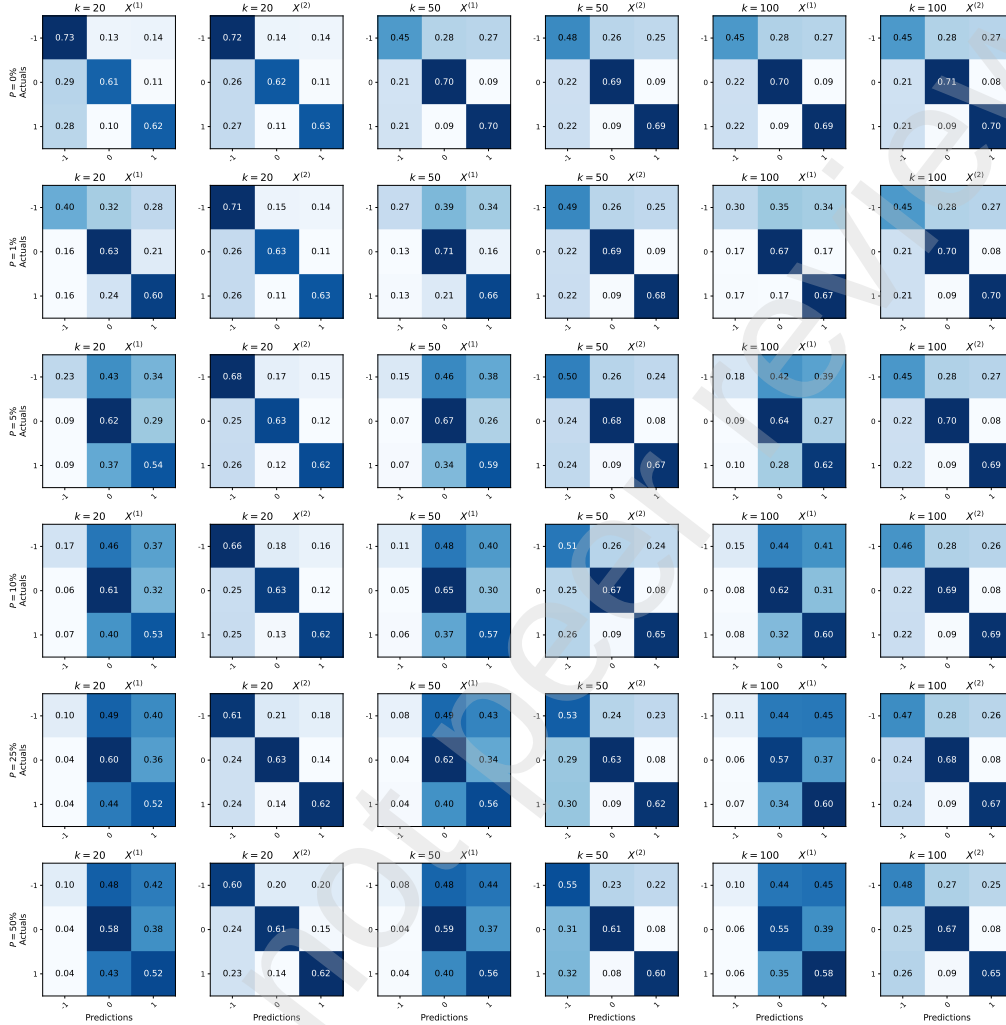


Figure 8: Confusion matrices for corresponding experimental results in Table 3.

leans towards generating near-zero positive predictions as noise levels escalate. Contrastingly, this degradation is far less severe using our proposed input representation, ranging from less than 0.5% ($P = 1\%$) to around 4% ($P = 50\%$). This finding supports our hypothesis regarding the stability and resilience of the proposed input data representation to emerging noises in the LOB data.

6. Conclusion

This paper introduced a novel approach to representing Limit Order Book data, leveraging the Ordered Fuzzy Numbers concept. The study was motivated by the challenges associated with the traditional level-based representation of LOB data, particularly its sensitivity to data perturbations. The proposed representation scheme aims to effectively capture the dynamic behaviour of LOBs while being considerably more robust to noise in the data. The proposed method was rigorously evaluated using year-long market data from the Warsaw Stock Exchange for the five largest companies.

Our results indicate several key insights. In a disturbance-free environment, the proposed OFN-based representation demonstrated comparable performance to traditional level-based schemes, slightly outperforming them in all evaluated metrics across various forecasting horizons. This confirms the efficacy of OFN in capturing critical information present in LOBs. One of the most compelling findings is the robustness of the proposed OFN representation to data perturbations. Traditional level-based representation schemes showed a significant decrease in prediction accuracy with increasing noise levels, dropping by up to 23% under certain conditions. In stark contrast, the OFN representation maintained a remarkably stable performance, with accuracy deteriorating by no more than 4%.

The OFN-based representation serves as a promising input method for machine learning models like DeepLOB and opens doors to other applications, such as defining liquidity measures or employing fuzzy time series methods. The results highlight the utility and robustness of the OFN-based representation, making it a promising alternative for future research and practical applications. Our future work would mainly focus on applying the proposed representation scheme to define new liquidity measures because capturing the aspects of liquidity, such as the depth of the market, was the main inspiration for developing the method. Moreover, future work may focus on exploring the applicability of this method to other types of financial instruments and investigating the benefits of integrating the proposed representation with other machine learning models.

Acknowledgement

We are most grateful to the Warsaw Stock Exchange (WSE), which approved releasing the open-access version of our data to share with the broader

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Appendix A. Model performance for individual stocks

This section showcases the remaining tables and confusion matrices that display the outcomes generated by the DeepLOB model for individual stocks.

Table A.4: Experiment results on KGHM for various perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	65.97	66.59	68.52	68.86	65.97	66.59	66.3	66.87
	50	62.92	62.85	62.6	62.77	62.92	62.85	62.68	62.76
	100	62.77	61.96	62.6	61.45	62.77	61.96	62.67	61.64
1%	20	48.68	66.1	49.68	68.12	48.68	66.1	44.83	66.37
	50	51.04	62.5	48.99	62.54	51.04	62.5	47.8	62.48
	100	50.27	61.77	50.27	61.28	50.27	61.77	48.61	61.46
5%	20	39.47	64.07	42.63	65.25	39.47	64.07	31.82	64.31
	50	40.82	61.06	39.9	61.64	40.82	61.06	33.73	61.25
	100	40.66	60.97	40.44	60.58	40.66	60.97	32.81	60.69
10%	20	38.04	61.93	40.67	62.53	38.04	61.93	30.04	62.06
	50	39.11	59.5	38.81	60.81	39.11	59.5	31.96	59.89
	100	39.02	60.01	38.88	59.83	39.02	60.01	30.55	59.79
25%	20	36.95	57.85	39.29	57.91	36.95	57.85	29.02	57.52
	50	37.61	55.75	37.68	59.28	37.61	55.75	30.66	56.39
	100	37.38	57.76	37.4	58.42	37.38	57.76	29.53	57.8
50%	20	36.38	55.91	38.01	55.7	36.38	55.91	28.98	55.24
	50	36.55	53.64	34.73	58.32	36.55	53.64	29.57	54.26
	100	36.17	56.32	35.91	57.74	36.17	56.32	28.91	56.55

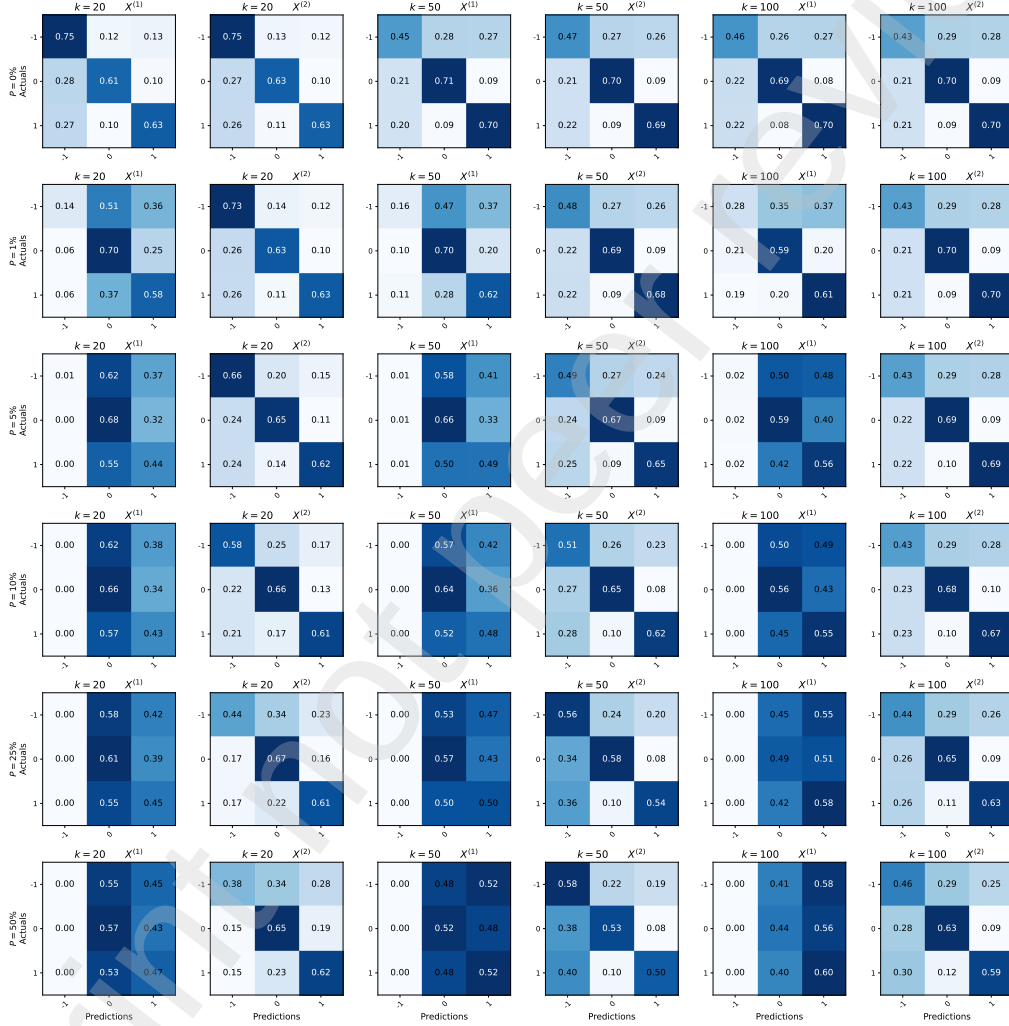


Figure A.9: Confusion matrices for corresponding experimental results in Table A.4.

Table A.5: Experiment results on PKNORLEN for various perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	64.09	65.2	66.6	66.61	64.09	65.2	64.6	65.54
	50	61.98	62.28	60.7	61.17	61.98	62.28	60.82	61.35
	100	61.2	61.86	60.01	60.75	61.2	61.86	59.62	60.77
1%	20	50.48	64.83	50.03	66.07	50.48	64.83	48.19	65.15
	50	51.11	62.02	49.87	61.02	51.11	62.02	45.89	61.24
	100	51.85	61.81	50.2	60.72	51.85	61.81	46.31	60.76
5%	20	41.85	63.56	43.46	64.37	41.85	63.56	35.92	63.8
	50	41.94	60.95	42.23	60.52	41.94	60.95	34.5	60.65
	100	43.5	61.51	42.51	60.5	43.5	61.51	35.78	60.64
10%	20	40.46	62.38	42.87	62.95	40.46	62.38	34.19	62.55
	50	40.2	59.75	41.31	60.14	40.2	59.75	32.77	59.86
	100	41.45	61.14	41.1	60.28	41.45	61.14	33.95	60.45
25%	20	39.31	60.28	41.47	60.67	39.31	60.28	32.83	60.35
	50	38.49	56.74	37.92	59.4	38.49	56.74	31.69	57.52
	100	38.8	60.0	37.51	59.72	38.8	60.0	31.57	59.76
50%	20	38.65	59.21	39.34	59.75	38.65	59.21	32.42	59.22
	50	37.39	54.71	35.48	58.96	37.39	54.71	31.47	55.75
	100	36.92	59.04	35.76	59.33	36.92	59.04	30.15	59.09

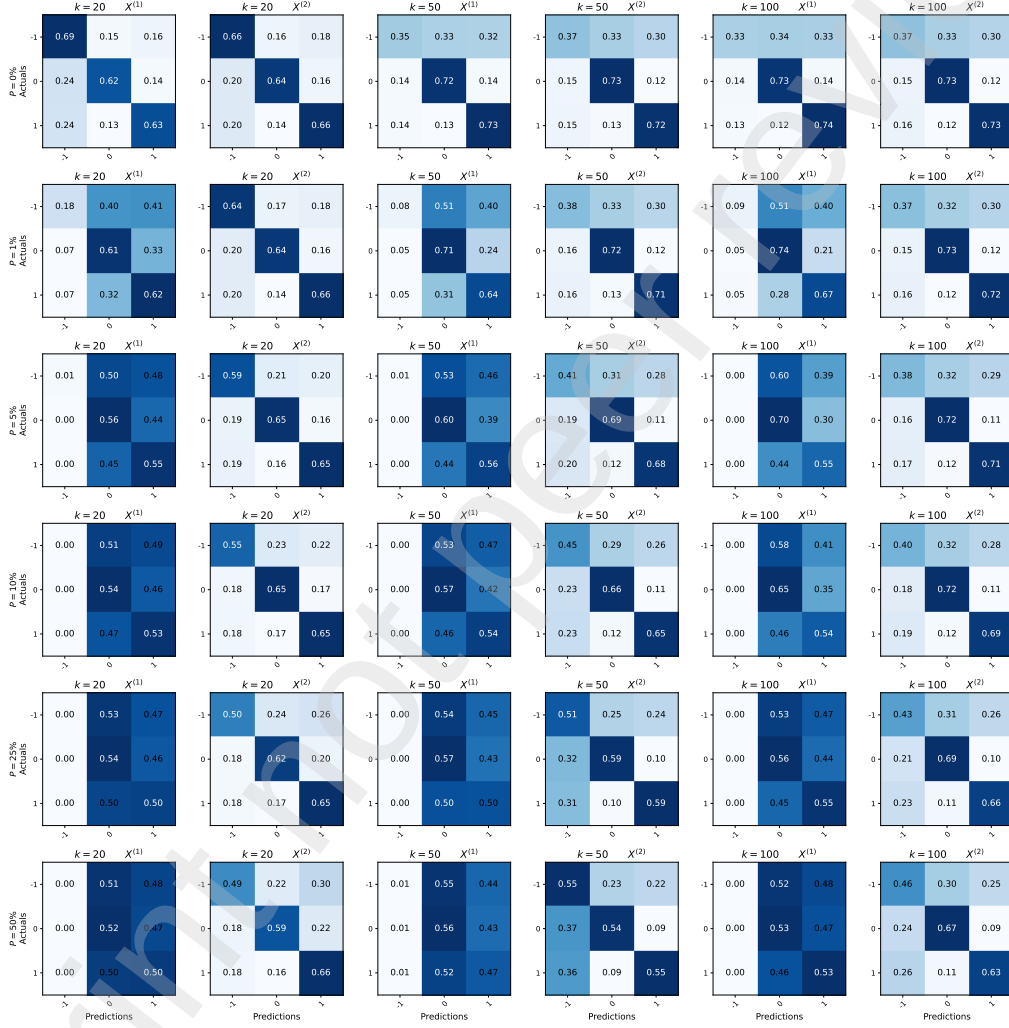


Figure A.10: Confusion matrices for corresponding experimental results in Table A.5.

Table A.6: Experiment results on PKOBP for various perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	64.75	65.53	68.18	68.23	64.75	65.53	64.96	65.74
	50	61.96	62.42	61.79	62.98	61.96	62.42	61.84	62.61
	100	61.33	62.15	61.12	62.2	61.33	62.15	61.15	62.15
1%	20	63.17	65.53	65.11	68.23	63.17	65.53	63.38	65.74
	50	60.51	62.42	59.97	62.97	60.51	62.42	59.97	62.6
	100	60.15	62.15	59.7	62.2	60.15	62.15	59.69	62.15
5%	20	57.56	65.53	57.44	68.22	57.56	65.53	57.35	65.74
	50	55.48	62.42	55.58	62.97	55.48	62.42	53.03	62.6
	100	56.07	62.16	55.7	62.2	56.07	62.16	54.43	62.15
10%	20	52.67	65.53	52.88	68.23	52.67	65.53	51.28	65.74
	50	51.43	62.42	53.0	62.97	51.43	62.42	47.06	62.6
	100	52.43	62.16	52.86	62.2	52.43	62.16	49.53	62.15
25%	20	47.27	65.53	48.67	68.23	47.27	65.53	43.78	65.74
	50	46.77	62.42	50.59	62.97	46.77	62.42	40.42	62.6
	100	47.73	62.16	49.9	62.2	47.73	62.16	42.97	62.15
50%	20	45.87	65.52	47.72	68.23	45.87	65.52	41.84	65.73
	50	45.34	62.41	49.85	62.97	45.34	62.41	38.63	62.6
	100	46.01	62.16	49.34	62.2	46.01	62.16	40.54	62.15

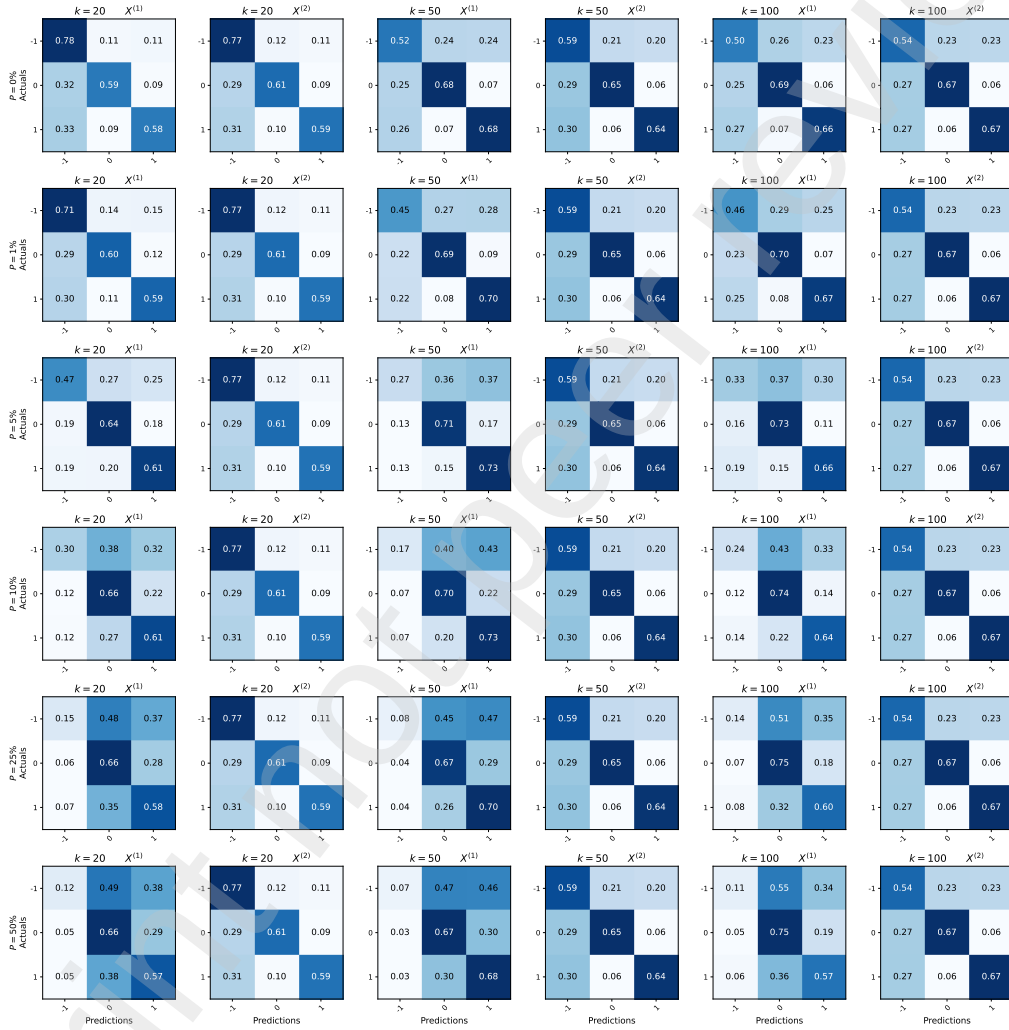


Figure A.11: Confusion matrices for corresponding experimental results in Table A.6.

Table A.7: Experiment results on PZU for various perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	63.71	65.11	66.29	68.03	63.71	65.11	64.05	65.4
	50	61.66	62.86	61.51	62.67	61.66	62.86	61.51	62.74
	100	61.5	63.09	61.63	62.65	61.5	63.09	61.48	62.79
1%	20	62.3	65.11	64.04	68.02	62.3	65.11	62.58	65.4
	50	60.66	62.86	60.35	62.66	60.66	62.86	60.37	62.74
	100	60.53	63.09	60.66	62.65	60.53	63.09	60.5	62.79
5%	20	57.87	65.11	58.02	68.02	57.87	65.11	57.88	65.4
	50	57.35	62.87	56.71	62.67	57.35	62.87	56.47	62.75
	100	57.5	63.09	57.52	62.65	57.5	63.09	57.28	62.79
10%	20	54.25	65.1	54.18	68.02	54.25	65.1	53.85	65.4
	50	54.84	62.87	54.22	62.67	54.84	62.87	53.38	62.75
	100	55.14	63.09	55.04	62.65	55.14	63.09	54.63	62.79
25%	20	49.89	65.11	50.31	68.02	49.89	65.11	48.67	65.4
	50	51.41	62.86	51.1	62.67	51.41	62.86	49.12	62.74
	100	51.79	63.09	51.58	62.66	51.79	63.09	50.66	62.8
50%	20	48.99	65.11	49.53	68.02	48.99	65.11	47.68	65.4
	50	50.35	62.87	50.11	62.67	50.35	62.87	47.89	62.75
	100	50.52	63.08	50.34	62.65	50.52	63.08	49.1	62.79

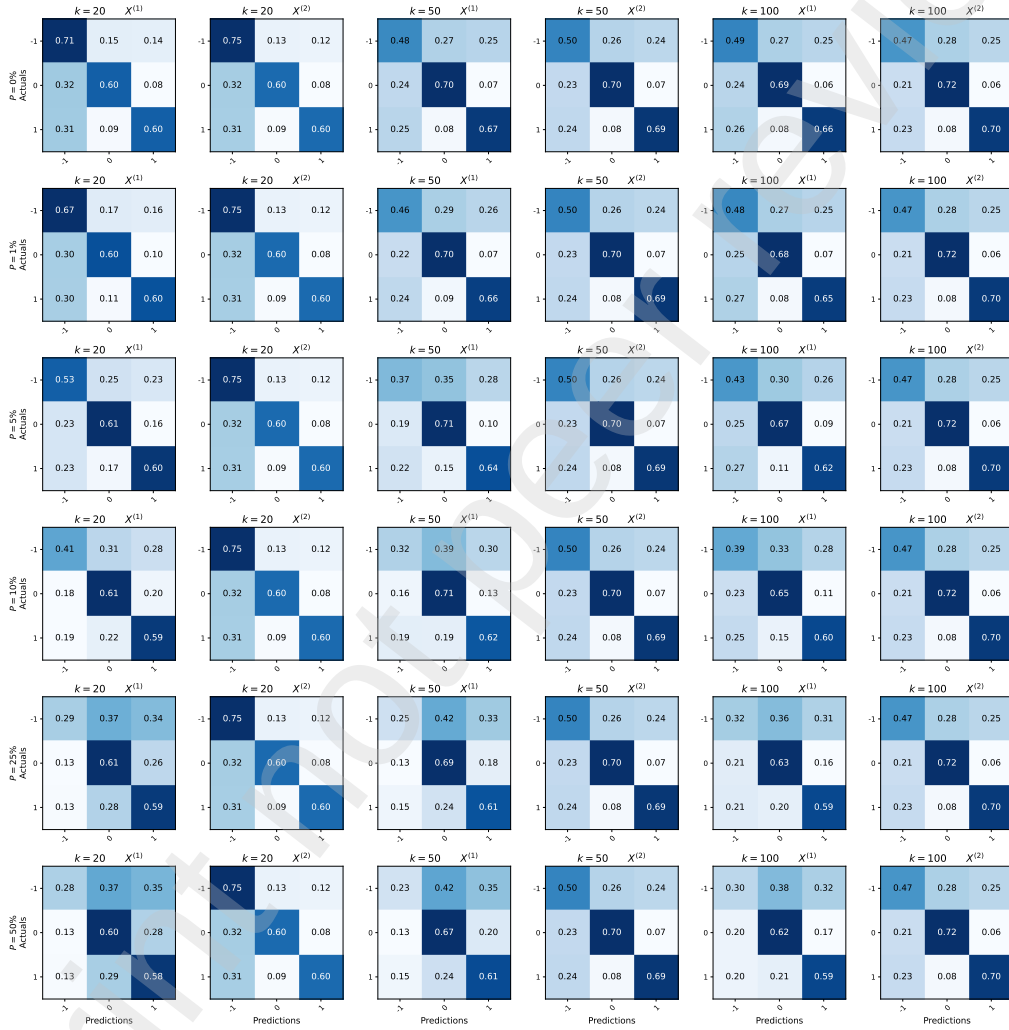


Figure A.12: Confusion matrices for corresponding experimental results in Table A.7.

Table A.8: Experiment results on PEKAO for various perturbation levels (P), prediction horizons (k) and types of inputs (level-based $X^{(1)}$ and proposed $X^{(2)}$).

P	k	Metrics (%)							
		Accuracy		Precision		Recall		F1-score	
		$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$	$X^{(1)}$	$X^{(2)}$
0%	20	64.0	64.35	65.62	65.36	64.0	64.35	64.33	64.59
	50	62.02	62.27	61.6	62.04	62.02	62.27	61.73	62.11
	100	60.64	60.54	60.8	60.02	60.64	60.54	60.65	60.17
1%	20	47.99	64.09	49.78	65.0	47.99	64.09	42.83	64.31
	50	52.13	62.16	52.55	61.93	52.13	62.16	45.34	62.0
	100	48.83	60.5	49.1	59.98	48.83	60.5	43.49	60.12
5%	20	39.28	63.05	45.42	63.67	39.28	63.05	31.68	63.22
	50	41.4	61.65	45.17	61.42	41.4	61.65	32.68	61.49
	100	38.88	60.32	40.3	59.79	38.88	60.32	30.1	59.93
10%	20	37.72	62.05	44.22	62.48	37.72	62.05	29.99	62.17
	50	39.02	61.2	42.06	61.0	39.02	61.2	30.13	61.06
	100	37.12	60.11	38.6	59.59	37.12	60.11	27.93	59.72
25%	20	36.56	60.23	40.34	60.57	36.56	60.23	28.88	60.31
	50	37.0	60.12	37.16	60.13	37.0	60.12	28.39	60.07
	100	35.33	59.61	36.34	59.15	35.33	59.61	26.18	59.24
50%	20	36.43	59.4	38.4	59.92	36.43	59.4	29.92	59.53
	50	36.12	59.48	36.16	59.72	36.12	59.48	29.53	59.5
	100	34.6	59.22	34.74	58.87	34.6	59.22	28.06	58.91

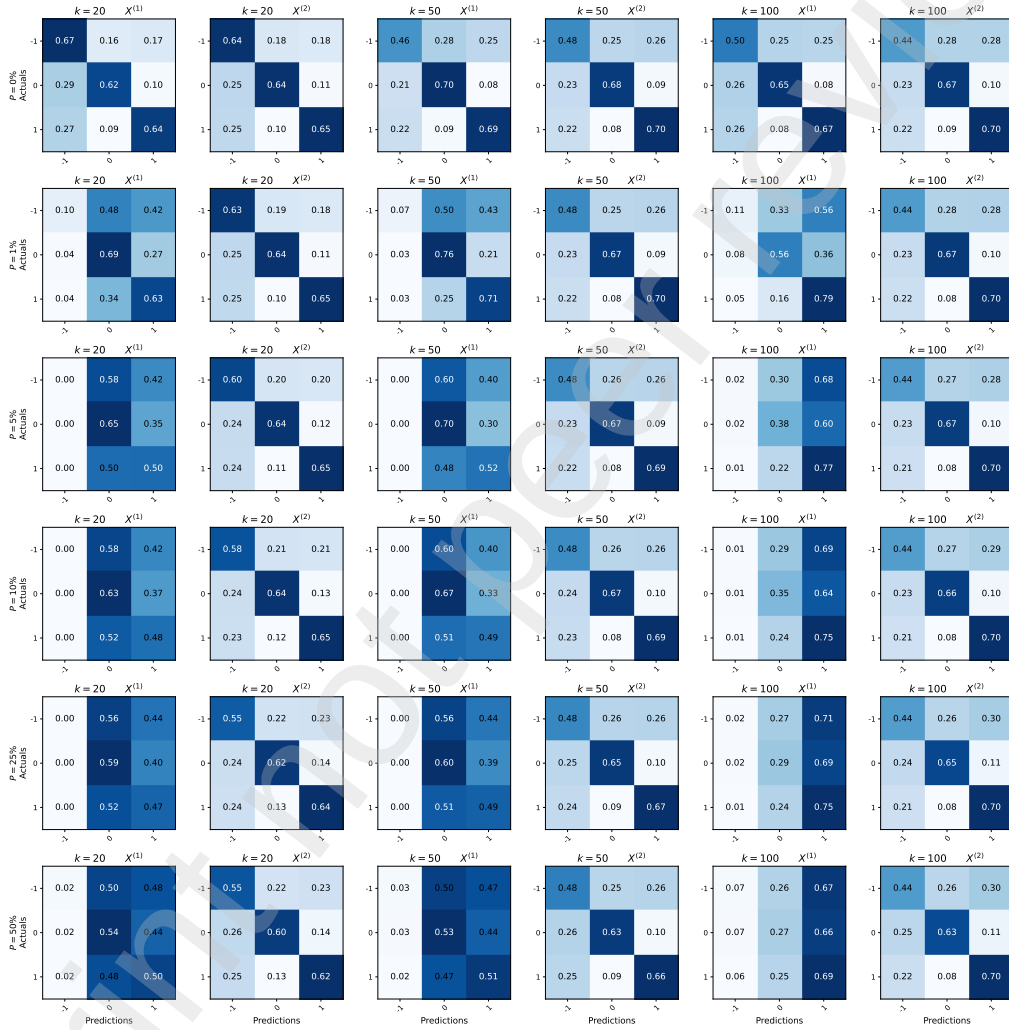


Figure A.13: Confusion matrices for corresponding experimental results in Table A.8.

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